

Solution Requirements (Functional & Non-functional)

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Team ID: Banka Akshay Kumar Reddy
Project Name: ShopEZ (E-Commerce Platform)

Functional Requirements (FR)

These outline the specific behaviors and features the system must support:

FR No.	Epic / Feature Area	Detailed Sub-Requirement
FR-1	User Management	Registration form, Login authentication, Stateless session handling, Role-based Access Control (distinguishing Customer vs. Admin).
FR-2	Product Catalog	Fetch all products from API, Filter by category/price, View detailed single-product pages with dynamic variants.
FR-3	Cart Operations	Add items to cart, Remove items, Automatically calculate total cost including potential taxes/discounts dynamically.
FR-4	Order Processing	Convert cart items into a finalized order, Clear user's cart post-purchase, Allow users to view their historical orders.
FR-5	Admin Panel	Secure dashboard to Add/Edit/Delete products, Update global store banners, View and manage lifecycle of user orders.

Non-Functional Requirements (NFR)

These define system attributes such as performance, usability, and security:

FR No.	Non-Functional Requirement	Description & Measurement
NFR-1	Usability & Accessibility	Must provide an intuitive, responsive UI that works flawlessly on mobile devices. Must include customized Light/Dark themes for user comfort.
NFR-2	Security & Privacy	Passwords must be hashed using Bcrypt (minimum 10 salt rounds). API endpoints must be protected using robust JWT stateless authentication.
NFR-3	Performance Optimization	Client-side routing via React Router v6 must ensure sub-second page transitions, providing a native app-like experience.
NFR-4	Database Scalability	Leveraging a NoSQL database (MongoDB) allows for a flexible schema, accommodating future product types and catalog expansion effortlessly.